

Matrix inequalities and norms

Spectral norms of independent entries with regular moment growth

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved**Topic:** Expected spectral norms of inhomogeneous random matrices

Rating rationale: The missing upper estimate must account for localized random fluctuations across all variance profiles. Even weighted random signs retain an iterated-logarithm gap. A sharp estimate would improve quantitative understanding of the largest singular value in random matrix models.

Statement

For a real random variable Z and $r > 0$, write $\|Z\|_{L_r} = (\mathbb{E}|Z|^r)^{1/r}$. Fix $\alpha \geq 1$. Let $X = (X_{ij}) \in \mathbb{R}^{n \times n}$ have independent mean-zero entries with finite absolute moments of every order, satisfying

$$\|X_{ij}\|_{L_{2r}} \leq \alpha \|X_{ij}\|_{L_r} \quad (1 \leq i, j \leq n, r \geq 1).$$

Set $[n] = \{1, \dots, n\}$ and

$$M(X) = \max_i \left(\sum_j \mathbb{E} X_{ij}^2 \right)^{1/2} + \max_j \left(\sum_i \mathbb{E} X_{ij}^2 \right)^{1/2},$$

$$D(X) = \max_{1 \leq k \leq n} \min_{\substack{I \subseteq [n] \\ |I| \leq k}} \sup_{\substack{s, t \in \mathbb{R}^n \\ \|s\|_2, \|t\|_2 \leq 1}} \left\| \sum_{\substack{i \in I \\ j \notin I}} X_{ij} s_i t_j \right\|_{L_{\ln(k+1)}}.$$

Conjecture (Łatała–Świątkowski). For every $\alpha \geq 1$ there is a constant $C_\alpha > 0$ such that, for every $n \geq 1$ and every such matrix,

$$\mathbb{E} \|X\|_2 \leq C_\alpha (M(X) + D(X)),$$

where $\|X\|_2$ is the spectral norm. The constant is independent of the dimension and entry laws. The same deterministic index set I deletes both rows and columns; the supremum is outside the random-variable moment. The definition includes $L_{\ln 2}$ when $k = 1$, using the displayed moment functional even though its exponent is below one.

The reverse inequality up to a constant depending only on α is proved, so the target is equivalent to the source's two-sided comparison. No identical-distribution or symmetry assumption is imposed on the entries.

Known cases and numerical significance

The source proves the Gaussian-mixture case under the same moment condition. Weighted signs $X_{ij} = a_{ij}\varepsilon_{ij}$, with independent fair $\varepsilon_{ij} \in \{-1, 1\}$, satisfy the condition with $\alpha = 1$ and form the source's earlier Conjecture 1.2. Latała proves this case when $a_{ij} \in \{0, 1\}$ and proves the general weighted-sign estimate with an additional factor of order $\log \log \log n$. Meller's 2026 result gives an iterated-logarithm loss for a wider class of symmetric entries; it does not remove that loss. These cases remain grouped in this entry.

The spectral norm is the largest singular value and measures the largest Euclidean amplification by a random matrix. The formula combines row and column variance scales with moments of bilinear forms after deleting a limited number of coordinates, accounting for concentration on small parts of a matrix that simpler variance-only bounds can miss.

References and status check

- R. Latała and W. Świątkowski, *Norms of randomized circulant matrices*, Electronic Journal of Probability 27 (2022), paper 80, 1–23. DOI; current arXiv v2, dated 2022-05-27. Conjecture 4.3 on preprint p.25, with condition (25) on p.24, states the target. Theorem 4.1 supplies the lower bound; Proposition 4.4 and the paragraph before it cover Gaussian mixtures. Conjecture 1.2 on p.2 is the weighted-sign special case.
- R. Latała, *On the spectral norm of Rademacher matrices*. DOI; current arXiv v2, dated 2025-08-18. Equations (1.2)–(1.3), Theorem 1.1, and Theorem 1.9 give the weighted-sign conjecture and the stated partial results. This paper uses the comparable truncated logarithm $\max\{1, \ln k\}$ in place of $\ln(k+1)$.
- R. Meller, *Spectral norm of matrices with independent entries up to polyloglog*, arXiv:2512.23673v2, dated 2026-01-29, introduction, equation (3), and Theorem 1.1. It explicitly identifies the remaining logarithmic gap and Conjecture 4.3.

On 2026-09-11, checked the original paper's current version, both later papers, and exact-title, author/conjecture, Rademacher spectral-norm, proof/counterexample and 2026 searches. No full resolution was found. Gaussian operator-norm results and the general Bernoulli-process theorem do not prove this displayed formula. This is a bounded literature check.